Chapter 04

Numerical Solutions of Boundary value Problems for Ordinary Differential Equations

Lesson 1

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1. Introduction

So far, we have considered **initial value problems (IVPs)**, where all the required conditions are specified at one point.

For example,

$$\frac{d^2y}{dx^2} + y = 0, \qquad y(0) = 1$$

Here, both conditions are given at $x = 0$.

A **boundary value problem (BVP)** is different because the conditions are usually specified at **different points** of the interval.

For example,

$$\frac{d^2y}{dx^2} + y = 0,$$

subject to

$$y(0) = 0, \qquad y\left(\frac{\pi}{2}\right) = 1.$$

Here, one boundary condition is given at $x = 0$, while the other is given at $x = \frac{\pi}{2}$.

Definition

A **boundary value problem** for an ordinary differential equation is a differential equation together with conditions imposed at two or more different points in the independent-variable interval.

- The simplest BVP consists of a 2nd order differential equation with boundary conditions given at the two ends of the domain.

A typical second-order BVP is

$$y'' = f(x, y, y'),\ a \le x \le b$$

with boundary conditions

$$y(a) = \alpha, y(b) = \beta$$

where

- a, b are the boundary points,
- α, β are specified boundary values.

In this section, we consider second order differential equations of the form,

$$y'' = f(x, y, y'),\ a \leq x \leq b \ \cdots (\mathbf{1})$$

with boundary conditions;

$$y(a) = \alpha, y(b) = \beta$$ (i.e. Dirichlet Boundary Conditions)

If the equation is second order differential equations,

Under the IVP, We need to determine the unknown initial slope

Example

Consider

$$y'' = f(x, y, y').$$

IVP

$$y(0) = 1, \qquad y'(0) = 2.$$

But under the BVP, just we need to know about boundary values,

BVP

$$y(0) = 1, \qquad y(1) = 2.$$

The conditions are specified at different points.

Main Numerical Method for BVPs

Shooting Method

The shooting method solves a boundary value problem (BVP) by ,

1. Converting it into an initial value problem (IVP) and

2. Adjusting the missing initial conditions iteratively until the boundary conditions are met.

BVP

$$y'' = f(x, y, y'),\ a \le x \le b \quad \cdots (\mathbf{1})$$

with boundary conditions;

$$y(a) = \alpha, y(b) = \beta$$ (i.e. Dirichlet Boundary Conditions)



IVP

$$y'' = f(x, y, y') \ , y(a) = \alpha, \qquad y'(a) = z = u$$

Steps

1. Start with an initial guess of u
2. Solve the IVP by marching from $x = a,\ to\ x = b$ (can use any method to solve, like Euler's, Rugge Kutta)
3. Stop if the solution agrees with $y(b) = \beta$
4. Otherwise, adjust u and try again.
5. And use linear interpolation method to find next guess.

Linear Interpolation Formula

$$Z = G_1 + \frac{G_2 - G_1}{R_2 - R_1}(D - R_1)$$

Guess 1 =G_1

Guess 2 =G_2 AND Result value we get under each guess y_n values take as $R_1\ and\ R_2$.

And D is the correct last boundary value $\mathrm{D} = y(b) = \beta$

Then again do the iterations

Example 01

Use the shooting method to solve the BVP.

$$\frac{d^2y}{dx^2} - 2y = 0, y(0) = 1.2, \qquad y(1) = 0.9 \; and \; h = 0.25$$

Use Euler's method to solve the corresponding IVP. Compute the value of y(0.25), y(0.5) and y(0.75) correct to 6 decimal places.

Example 02

Use the shooting method to solve the BVP.

$$\frac{d^2y}{dx^2} = x + y, y(0) = 1, \qquad y(1) = 0 \; and \; h = 0.25$$

Use Euler's method to solve the corresponding IVP. Compute the value of y(0.25), y(0.5) and y(0.75) correct to 6 decimal places.